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Real Time Red Palm Weevil Detection in Coconut Tree via Multimodal Sensor Data Based Deep Learning Network

Siron Anita Susan T · Manjunathan Alagarsamy · Sumathi Gurusamy ·
Alima Beevi Azizur Rahuman · V. Ramya · L. K Sowmya Sundari

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Abstract

Red Palm Weevil (RPW) infestation poses a significant threat to coconut trees leading to severe damage and economic losses. However, existing RPW detection techniques have low sensitivity to early infestations as internal damage by larvae is not externally visible, hindering timely intervention. In this work, a novel deep learning-based RED-BIYO is proposed for detecting coconut tree damages using multimodal data namely stem image and audio signal. The stem images are gathered by an image capture device and audio signals are captured by a coconut microphone sensor. The stem image is pre-processed by Savitzky-Golay (SavG) filter to remove the noise from the images. The YOLOv9 model leverages a ResNeSt-based split-attention backbone and a Feature Pyramid Network (FPN) to effectively extract detailed multi-scale features from pre-processed images for enabling precise detection of Damaged Tree (DT) and No Damaged Tree (NDT). Simultaneously, the acquired audio signals are pre-processed using short-time framing with Hamming windowing and STFT-based spectrogram extraction. The resulting spectrograms are analyzed using the Spiking Convolutional Neural Network based Bidirectional Gated Recurrent Unit (SCNN-BiGRU) for RPW detection. SCNN extracts spatial and frequency features, while BiGRU captures temporal patterns in both forward and backward directions. The multi model data are fused in fuzzy rules to assess coconut tree health and notify farmers via an IoT-enabled Blynk app. The proposed RED-BIYO achieves the detection accuracy (AC) of 99.54% and Matthews Correlation Coefficient (MCC) of 97.93%. The proposed RED-BIYO model increases the overall accuracy by 2.75%, 9.38%, 2.61%, 22.16%, 8.54% and 4.39% of YOLOv5, ResNet50, Mask R-CNN, InceptionResNetV2, YOLOv3 and MIN-SVM model respectively. In future, RED-BIYO will focus on optimizing the model for low-power edge deployment to enable large-scale, real-time RPW monitoring.

Keywords Red palm weevil · Multimodal data · Deep learning · Savitzky-Golay filter · Blynk app

1 Introduction

Coconut trees are widely cultivated in tropical regions around the world and are an important source of livelihood for many people. However, these trees are disposed to a variety of diseases, it reduces their productivity and even cause death [1]. Effective disease management and prevention of further spread require early detection and



diagnosis of these diseases [2]. The traditional method of identifying diseases involves expert visual inspection of trees, which is time-consuming and vulnerable to human error. The coconut tree diseases need to be detected and diagnosed quickly by farmers and agricultural experts through a fast and accurate disease identification system [3]. In recent years, deep learning (DL) techniques have revolutionized the field of image recognition and classification [4]. Healthcare, finance, agriculture, and other fields have shown considerable success with DL algorithms [5].

Smart precision agriculture integrates existing information and wireless transmission networks to improve automated, consistent agricultural operations and techniques [6]. In real-time farming applications, the Internet of Things (IoT) can detect insect pests, manage irrigation water accurately, or detect crop diseases [7]. The IoT primarily depends on wireless sensor networks (WSNs) to detect and collect environmental factors like humidity and soil moisture. The collected data might then be processed and saved for use in water irrigation pumps and to support expert choices [8]. Current environmental and agricultural data are analysed using artificial intelligence systems such as DL [9] and ML [10] methodologies.

Precision agriculture is based on crop health observation to detect the infectious status of insect pests in agricultural regions [11]. The use of conventional approaches and manual identification of insect pests has proven ineffective, time-consuming, and expensive [12]. As a result, faster detection of plant pests is more important for agriculturalists to use suitable pesticides and minimize crop loss [13, 14]. However, existing RPW detection techniques have low sensitivity to early infestations as internal damage by larvae is not externally visible, hindering timely intervention. In this work, a novel IoT-based RED-BIYO is proposed for detecting coconut tree damages using multimodal data namely audio signal and stem image. The main contribution of the research are as follows,

- Stem images are captured using a high-resolution image acquisition device to record visible damage symptoms on the coconut tree surface. Simultaneously, acoustic signals are recorded through a coconut-mounted microphone sensor to capture internal larval activity associated with RPW infestation.
- Stem images are analyzed using the YOLOv9 detection framework integrated with a ResNeSt-based split-attention backbone and FPN that enables effective multi-scale feature extraction and precise localization for accurate detection of DT and NDT conditions.
- Audio spectrogram are processed using the SCNN-BiGRU architecture, where the SCNN captures discriminative spatial–frequency patterns and the BiGRU models forward and backward temporal dependencies to achieve robust RPW classification.
- The multimodal outputs are fused using a fuzzy logic-based decision system to generate a comprehensive assessment of coconut tree health and the final decision is transmitted to farmers through an IoT-enabled Blynk application for timely monitoring and intervention.

The rest of this paper is arranged as follows, the literature survey is summarized in Sect. 2, Proposed RED-BIYO methodology illustrates in Sect. 3. Section 4 represents the experimental result and discussion. Section 5 signifies the conclusion and future work of the proposed framework.

2 Literature Survey

The DL techniques in the field of agriculture has gained important attention in recent years due to their capability to solve complex difficulties related to crop management and disease diagnosis. Several studies have explored the use of DL algorithms for the identification of plant diseases including coconut tree diseases.

In 2023 Wijethunga, C.D., et al., [15] suggested three models for several facets of coconut farming: prediction of coconut plant development, ideal classification for coconut growth, and soil classification. CNN, logistic regression (LR), and multivariate polynomial regression were among the machine learning (ML) methods employed to ensure that each model produced accurate results. The yellowing of leaves was detected using the YOLOv5 object detection method, which had an average precision of 96.87%.

In 2021 Iqbal, M.S., et al., [16] suggested an OpenAerialMap and WeRobotics which was utilized in the World Bank's AI competition to support the identification of coconut trees and the detection. The segmentation was performed by using the Mask R-CNN utilizing ResNet50 and ResNet101. The average accuracy in recognizing coconut trees was 91%.

In 2025 Vidhanaarachchi, S., et al., [17] suggested a TL-based CNN and Mask R-CNN to detect WCWLD and CCI in their early phases. The approach successfully segments disease areas, allowing for accurate localization and prompt intervention techniques. In addition, the suggested WCWLD illness severity diagnosis approach has a 97% accuracy rate.

In 2025 Goswami, A. and Kirit, D.D [18]., suggested a ML technology to predict the health of coconut trees. These coconut trees have high commercial worth but are also environmentally significant, and they are susceptible to a variety of illnesses and pests that can significantly lower productivity. The study examines multiple ML algorithms using data from various satellite images, drone-based sensors, and field data, such as CNN, Random Forest (RF), and SVM.

In 2021 Singh, P., et al., [19] suggested a deep 2D-Convolutional Neural Network (CNN) for detecting coconut tree disease. The custom-designed 2D-CNN was trained to forecast disease and pest infections. Furthermore, InceptionResNetV2 and MobileNet had classification accuracy of 81.48% and 82.10%, respectively, with Cohen's Kappa values of 0.77 and 0.74.

In 2022 Karunarathna, et al., [20] suggested a DL in conjunction by unmanned aerial vehicle (UAV) remote sensing to detect and map the spatial distribution of individual coconut trees. UAV remote sensing was used to capture high-resolution, true-color images of coconut trees at the Mahayaya Coconut Model Plantation in Sri Lanka. The findings indicate that the YOLOv3 model's validation accuracy average is 91.7%.

In 2024 Mendel, Z., et al., [21] proposed a Sensor-Based Infestation Mapping and Risk Assessment (SBIMRA) to locate weevil-infested palms and concentrating management efforts on colonized plants. To explore curative treatments, attack probability based on infestation history, and the impact of sensor-based management on infestation incidence reduction, sensor data are converted into damage index values. This is quite successful, providing a cost-effective alternative to preventative treatment dependence.

In 2022 Alsanea, M., et al., [22] suggested a region-based convolutional neural network (R-CNN) algorithm for identifying and distinguishing RPW. To determine the position of the RPW in a picture, an R-CNN algorithm was utilized to create bounding boxes around it. The development of an efficient, low-cost RPW detection and classification system for palm trees in the Al-Qassim region, which contains thousands of farms.

In 2021 Kurdi, H., et al., [23] suggested a Naive Bayes, KSTAR, AdaBoost, Bagging, PART, J48, MLP, SVM, RF, and LR for early-stage RPW infection. A genuine RPW dataset was used to assess the detection algorithms' accuracy, precision, recall, and F1 performance. The experimental findings shown that RPW infections may be predicted with an accuracy of 93%.

In 2023 Megalingam, R.K., et al., [24] suggested a DL based MIN-SVM for detecting coconut tree illness. The technique is based on three morphological factors: direction, disposition, and height. The MINSVM surpassed the Visual Geometry Group of 91.90%, InceptionNet of 81.66%, and ResNet of 71.95% with an accuracy of 95.35%. The Comparative Analysis of Existing Coconut Disease and RPW Detection Methods shown in Table 1.

According to the literature review, various DL, ML, and IoT framework were utilized for RPW monitoring and stem disease detection. However, existing RPW detection techniques have low sensitivity to early infestations as internal damage by larvae is not externally visible, hindering timely intervention. The main goal of the research is to develop a novel DL-based RED-BIYO model for detecting RPW infestations in coconut trees using multimodal data.

Table 1 Comparative Analysis of Existing Coconut Disease and RPW Detection Methods

Year	Author	Proposed Method	Advantages	Disadvantages
2023	Wijethunga et al., [15]	CNN, LR, Polynomial Regression, YOLOv5	Multi-task modeling	Focuses mainly on visible symptoms
2021	Iqbal et al., [16]	Mask R-CNN	Accurate aerial segmentation for large plantations	Requires high-resolution aerial imagery
2025	Vidhanaarachchi et al., [17]	TL-based CNN+Mask R-CNN	Early-stage disease detection and good localization	limited generalization discussion
2025	Goswami & Kirit [18]	CNN, RF, SVM	Multi-source data	Depends on remote sensing availability
2021	Singh et al., [19]	Custom 2D-CNN, Inception-ResNetV2, MobileNet	Lightweight architectures	Lower accuracy
2022	Karunarathna et al., [20]	YOLOv3 with UAV imagery	Effective tree mapping using UAV	Only spatial mapping
2024	Mendel et al., [21]	SBIMRA	Sensor-driven infestation risk modeling	No deep learning-based classification
2022	Alsanea et al., [22]	R-CNN for RPW detection	Bounding box localization of RPW	High computational cost
2021	Kurdi et al., [23]	NB, SVM, RF, AdaBoost, etc.	Comparative ML analysis	Lacks deep feature extraction
2023	Megalingam et al., [24]	MIN-SVM	Morphological feature-based classification	Depends on handcrafted features

3 Proposed Red-Biyo Methodology

In this research, a novel RED-BIYO is proposed for detecting RPW in coconut trees using multimodal data. Figure 1 shows the proposed RED-BIYO methodology.

3.1 Savitzky Golay Filter for Image Pre-Processing

In this section, the input coconut tree images are denoised using the Savitzky-Golay (SavG) filter to reduce noise artifacts. This filter effectively minimizes irrelevant variations while preserving essential structural details of the image. It operates by fitting a low-order polynomial to successive subsets of the data using the linear least squares method, and the fitted polynomials are then convolved with the signal. Mathematically, the image is represented using a set of convolution coefficients C_i and data points $n \{X_j, Y_j\}$ with the filtered output expressed as:

$$Y_j = \sum_{i=-(m-1)/2}^{i=(m-1)/2} C_i y_{j+i} \quad \frac{m-1}{2} \leq j \leq n - \frac{m-1}{2} \quad (1)$$

Where,

- Y_j -filtered output value at position j ,
- C_i -Convolution coefficient at index i ,
- y_{j+i} - original noisy input value at position $j+i$ in the image data,
- m -frame size,
- n - total number of data points.

To apply the SavG filter, three hyperparameters must be defined: the noisy image Y , the frame size m , and the polynomial order C . These hyperparameters directly affect the smoothing behaviour larger frame sizes yield stronger smoothing, while higher polynomial orders can preserve finer details but risk overfitting noise. In this study, the values of m and C were selected using a combination of trial-and-error and prior experience based on the image signal-to-noise ratio (SNR). Filtered outputs were evaluated by computing the correlation (COR) between the filtered and original images:

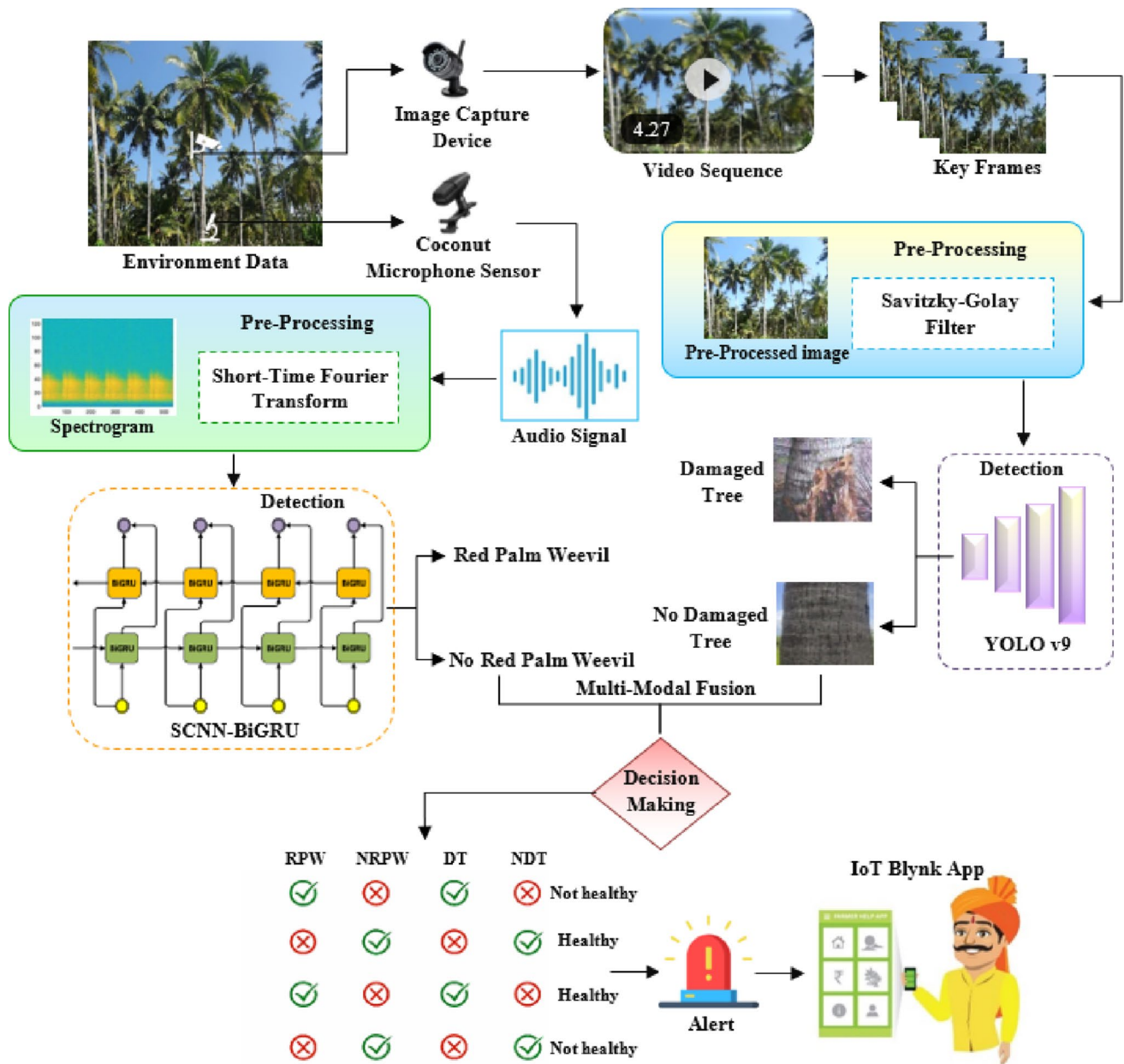


Fig. 1 Proposed RED-BIYO methodology

$$COR = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 (Y_i - \bar{Y})^2}} \quad (2)$$

Where,

- X_i - i-th pixel value from the original synthetic image dataset X,
- Y_i -i-th pixel value from the filtered image dataset Y,
- $\bar{X}$ -mean of the pixel values in the original image dataset X,
- $\bar{Y}$ -mean of the pixel values in the filtered image dataset Y,

- n - total number of pixels in the image.

Optimal filtering is considered achieved when COR approaches 1.00. Explicitly defining and evaluating these parameters ensures the reproducibility of the filtering process and enables others to replicate or adapt the approach to similar datasets.

3.2 Short-Time Fourier Transform Based Audio Pre-Processing

The recorded acoustic signal is segmented and transformed into a time–frequency representation using Short-Time Fourier Transform (STFT). The windowed signal is expressed as:

$$x_w(n) = x(n) \cdot w(n), \quad 0 \leq n \leq N - 1 \quad (3)$$

Where,

- $x_w(n)$ -windowed acoustic signal,
- $x(n)$ -original discrete time acoustic signal,
- $w(n)$ -hamming window function,
- N -frame length.

The hamming window is defined as:

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N - 1}\right) \quad (4)$$

The Short-Time Fourier Transform (STFT) of the windowed signal is computed using a 512-point FFT as:

$$STFT(m, k) = \sum_{n=0}^{N-1} x_w(n) e^{-j2\pi kn/N} \quad (5)$$

Where,

- $STFT(m, k)$ - time–frequency coefficient at frame m and frequency bin k ,
- $x_w(n)$ - windowed signal,
- N -FFT length,
- k -frequency bin index.

The magnitude spectrogram is obtained as:

$$S(m, k) = |STFT(m, k)| \quad (6)$$

Where,

- $STFT(m, k)$ - magnitude spectrogram.

To normalize the spectrogram values, min–max normalization is applied:

$$S_{norm}(m, k) = \frac{S(m, k) - S_{min}}{S_{max} - S_{min}} \quad (7)$$

Where,

- $S_{norm}(m, k)$ -normalized spectrogram value,
- S_{min} -minimum spectrogram value,
- S_{max} -maximum spectrogram value.

This preprocessing procedure enhances discriminative acoustic features while maintaining temporal and frequency resolution required for RPW detection.

3.3 Augmentation

Data augmentation techniques were employed to improve model generalization and reduce overfitting. These techniques were applied to both stem images and acoustic signals to increase data diversity and enhance robustness. For image augmentation, the input images were resized and subjected to random horizontal flipping ($p=0.5$), rotation between -20° and $+20^\circ$ and zooming up to 25% to introduce geometric variability while preserving semantic structure. For audio augmentation, controlled transformations such as additive Gaussian noise, polarity inversion, pitch scaling, time shifting, time stretching (0.9–1.1 factor) and random gain adjustment were applied to the raw acoustic signals to simulate real plantation recording conditions. These augmentation strategies increase dataset diversity, improve robustness against environmental variations and enhance the overall performance of the proposed RED-BIYO framework.

3.4 Coconut Tree Damaged Detection Using YOLOv9

The YOLOv9 network is employed to detect DT and NDT from noise-free images by utilizing its advanced backbone and feature fusion modules. It efficiently captures multi-scale features from the input, enabling the identification of fine visual cues such as boreholes, discoloration, and subtle texture irregularities associated with early-stage damage. It contains three main components: Backbone, Neck, and Head, which extract features and then produce accurate predictions, including bounding boxes, objectness scores, and class labels. Figure 2 shows the YOLOv9 structure.

3.4.1 Backbone

Initial feature extraction from input images and progressive feature representation enhancement while preserving computing efficiency are handled by the YOLO-V9 backbone. The initial block ensures smooth integration and data flow by effectively sending input photos to the Auxiliary and Backbone sections without modifying data. The convolutional blocks use the SiLU activation function, Batch Normalization, and 2D convolutional layers to extract complex features from the denoised image. During convolutional procedures, AutoPad dynamically modifies padding to maintain spatial dimensions. The inspiration from ResNeSt split-attention mechanisms, this setup improves detection accuracy while maintaining computational efficiency. The split-attention blocks divide feature maps into groups and subgroups, allowing attention-weighted fusion within each group. This selective feature aggregation refines semantic context and enhances representational power, especially for small or complex targets. The backbone incorporates Split-Attention Bottlenecks to facilitate down-sampling and effective feature transformation.

$$I \in \mathbb{M}^{H \times W \times 3} \quad (8)$$

Where,

- i -input image,
- H -Height of the image,
- W -Width of the image,
- M -Represents the set of all image pixel values.

To extract hierarchical features, the image is fed into a series of convolutional layers and split-attention blocks. The standard convolution operation is defined as:

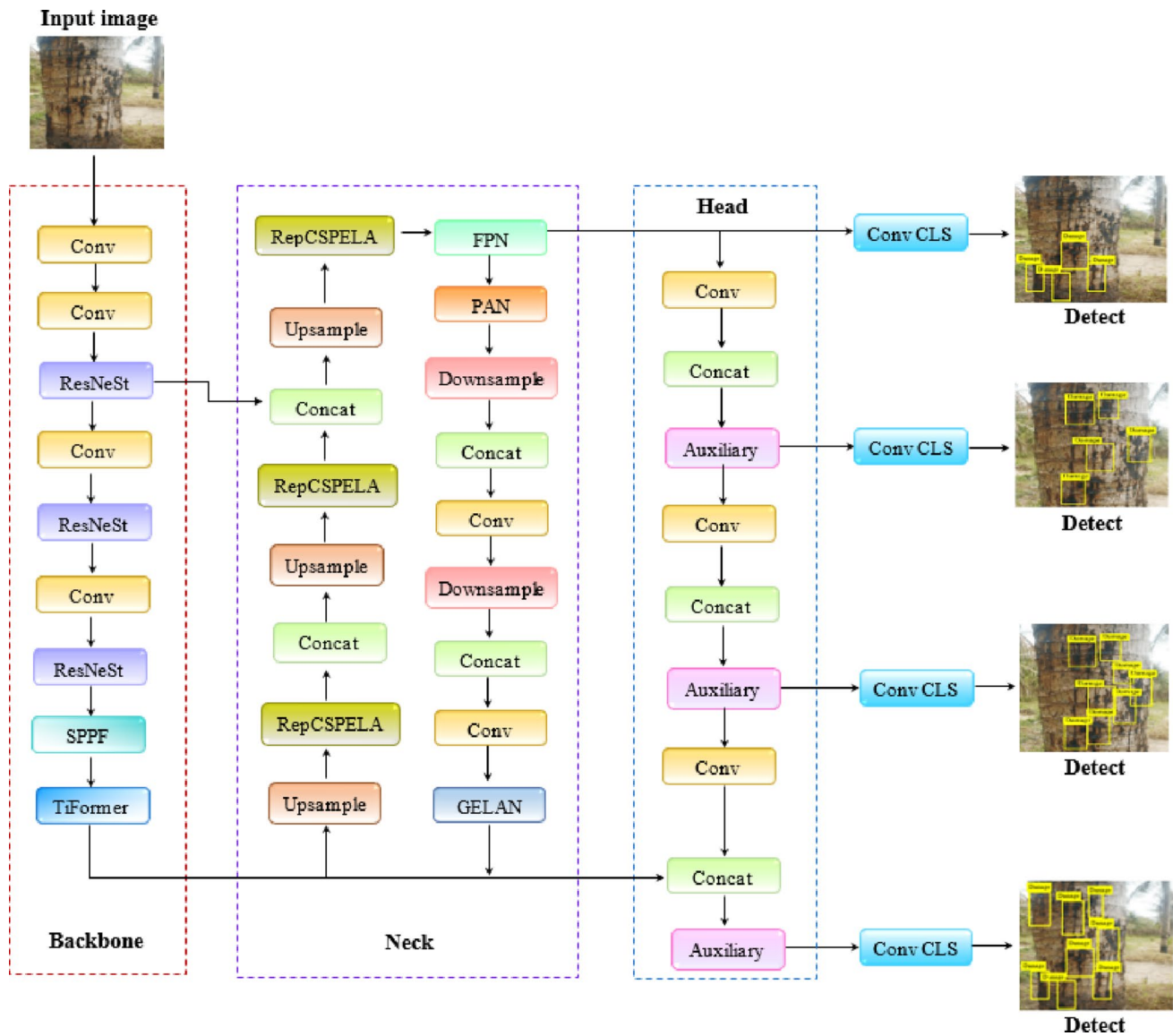


Fig. 2 YOLOv9 architecture

$$\text{Conv}(I, N, c, b) = \text{SiLU}(\text{BatchNorm}(I, \mathcal{K})) \quad (9)$$

Where,

- I-input feature map,
- N-Number of filters,
- c-stride,
- b-padding,
- $\mathcal{K}$ -convolutional kernel,
- BatchNorm- batch normalization,
- SiLU- activation function.

To capture deeper semantic features with enhanced context, ResNeSt introduces the following residual split-attention mechanism:

$$ResNeSt(X) = X + SplitAttn(Conv(Conv(X, \mathcal{K}_1, c_1, b_1), \mathcal{K}_2, c_2, b_2)) \quad (10)$$

Where,

- X - Input feature map to the ResNeSt block,
- $Conv()$ -Standard 2D convolution operation applied to feature maps,
- $SplitAttn$ - Split-attention fusion across feature groups within each block,
- $\mathcal{K}_1, \mathcal{K}_2$ - kernels of the convolutional layers,
- c_1, c_2 -Output feature maps,
- b_1, b_2 -Bias terms,
- $ResNeSt(X)$ - Final output of the block combining the original input (X) with the attention-enhanced feature representation through a residual connection.

By integrating ResNeSt into the YOLOv9 backbone, the model benefits from both residual learning and attention-based feature refinement, leading to better localization and recognition of objects with varied scale and appearance, all while optimizing computational cost.

3.4.2 Neck

The Neck module extract features for precise object detection obtained from the Backbone. It enhances feature representation by combining feature maps from multiple levels of the Backbone. The FPN integrates feature maps from several scales. In order to match previous Backbone outputs, the Up-Sampling Layer raises the resolution of feature maps from the CSPELAN block, enabling thorough feature integration. The Pre-serves fine-grained information essential for precise object localization by combining up-sampled feature maps with corresponding resolution feature maps from previous stages. By implementing more convolutional and Rep in Bottleneck blocks, features are refined guaranteeing that aggregated features are optimum for future detection tasks. Feature maps from various scales are combined in the FPN:

$$\mathcal{P}_l = UpSample(C_{l+1}) + C \quad (11)$$

Where,

- $UpSample$ - up-sampling,
- $\mathcal{P}_l$ - feature map at level l ,
- C_{l+1} - feature map from the layer,

The Path Aggregation Network (PAN) combines feature maps in both top-down and bottom-up paths to improve the feature pyramid:

$$U_l = Conv(Concat(\mathcal{P}_{l-1}, DownSample(\mathcal{P}_l))), \quad (12)$$

Where,

- U_l -output feature map at the level of l ,
- $Conv()$ - convolution operation,
- $concat$ -concatenation operation,
- $\mathcal{P}_{l-1}$ -Feature map from the previous level ($l-1$),
- $\mathcal{P}_l$ -feature map at the current level (l),
- $DownSample$ - downsampling operation.

The outputs are guaranteed to fall inside a suitable range by the sigmoid function. The definition of the objectness score is:

$$o = \sigma(t_c) \quad (13)$$

Where,

- o - objectness score.
- σ - sigmoid activation function,
- t_c -raw class score.

3.4.3 Head

The Head of YOLOv9 predicts bounding boxes and class probabilities for Auxiliary modules to complete object detection. These blocks leverage enhanced characteristics from the Neck module to produce accurate bounding box coordinates and class predictions, and are specifically designed to recognize fall motions at various scales. To guarantee precise predictions, YOLOv8 is trained using a loss function that incorporates several elements. The total loss is then defined as:

$$l = \lambda_{loc}l_{loc} + \lambda_{obj}l_{obj} + \lambda_{cls}l_{cls} \quad (14)$$

where,

- l - Total loss,
- λ_{loc} -localization loss,
- λ_{obj} -objectness loss,
- λ_{cls} -classification loss.

It combines strong detection and feature extraction, utilizing elements such as the Auxiliary section to enhance gradient flow and training resilience. Finally, YOLOv9 ensures robust real-time detection with high accuracy and efficiency for smart agricultural monitoring.

Table 2 shows the training hyperparameters configured for the YOLOv9 model used for coconut stem damage detection. The input images were resized to 640×640 pixels to maintain a balance between detection accuracy and computational efficiency. Auto-anchor with k-means optimization was employed to adapt anchor boxes according to the object size distribution in the dataset. The model was trained for 100 epochs with a batch size of 16 to ensure stable learning and effective convergence. Stochastic Gradient Descent (SGD) with a momentum value of 0.937 was used as the optimizer to accelerate gradient updates and reduce oscillations during training. Furthermore, a cosine annealing learning rate schedule with an initial learning rate of 0.01 was adopted to gradually reduce the learning rate for improving convergence stability and preventing overfitting.

Table 2 YOLOv9 Training Hyperparameters

Parameter	Value
Input Size	640×640
Anchors	Auto-anchor
Training Epochs	100
Batch Size	16
Optimizer	SGD with momentum
Learning Rate	0.01

3.5 Red Palm Weevil Sound Detection using Spiking CNN BiGRU (SCNN-BiGRU) Model

The proposed SCNN-BiGRU model is designed to detect RPW from acoustic signals collected using coconut tree monitoring sensors. This hybrid architecture integrates the SCNN with the temporal sequence learning strength of a BiGRU. Figure 3 illustrates the overall SCNN-BiGRU architecture.

The input to the model is a normalized spectrogram of size 128×128 obtained from the preprocessing stage. These continuous spectral representations are converted into spike-based tensors using a rate coding mechanism over $T=10$ discrete time steps. Each pixel intensity is treated as a firing probability, generating spike trains suitable for spiking neural computation. The spike-encoded spectrogram is represented as a tensor, $S(f) \in \mathbb{R}^{u \times v \times f}$ where (u, v) denote spatial coordinates and f represents the temporal axis. Each spatial location receives a spike train $s_{u,v}(f)$. A 3D convolutional kernel is applied to this tensor to extract meaningful spatial-temporal patterns with overlapping receptive fields. The spike-based feature extraction is mathematically expressed as:

$$a_{u,v}(f) = s_{u,v}(f) * \sigma(f) \quad (15)$$

Where,

- $a_{u,v}(f)$ -Weighted spike activity at spatial location (u, v) and time frame f ,
- $s_{u,v}(f)$ - spike at location (u, v) and time f .
- $\sigma(f)$ - Synaptic response function at time f .

$$u_{j,k}(f) = \sum_{m=1}^K \sum_{n=1}^K W_{m,n} a_{j+m-1, k+n-1}(f) + (s_{j,k}(f) * v(f)) \quad (16)$$

Where,

- $u_{j,k}(f)$ -Membrane potential at neuron (j, k) and time f ,
- K -kernel size,
- $W_{m,n}$ -Weight of the convolutional kernel at position (m, n) ,
- $s_{j,k}(f)$ -Local spike input at position (j, k) and time f ,
- $v(f)$ - Bias or modulation term at time f .

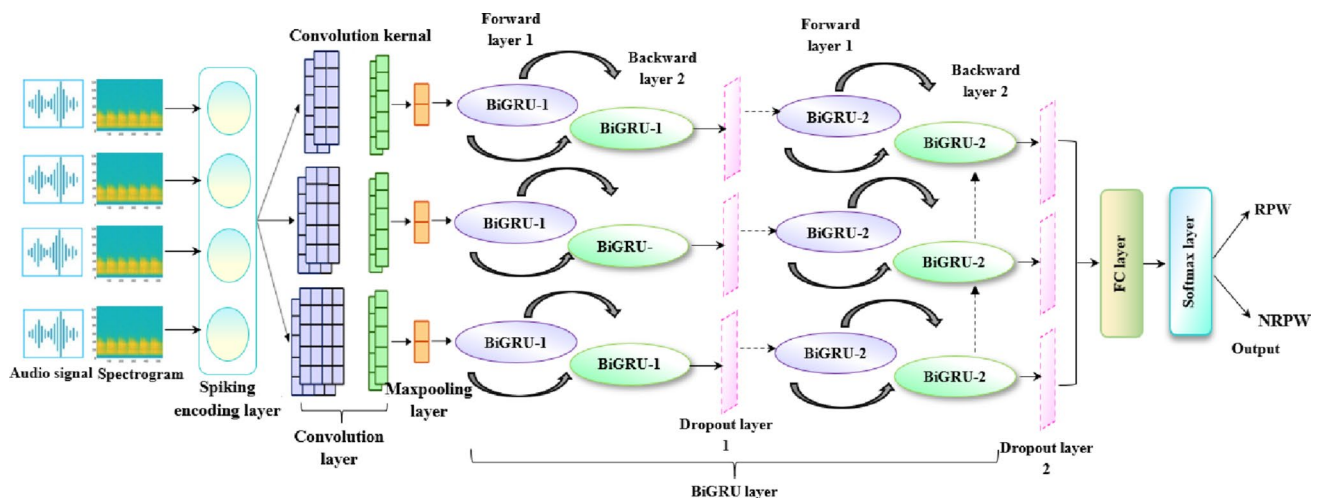


Fig. 3 Architecture of SCNN-BiGRU model

$$s_{j,k}(f) = 1 \& u_{j,k}(f) = 0 \quad \text{when } u_{j,k}(f) \geq V_{thr} \quad (17)$$

Where,

- $s_{j,k}(f)$ -Output spike,
- $u_{j,k}(f)$ -Membrane potential at position (j, k) ,.
- V_{thr} -Firing threshold,

The SCNN layers employ a Leaky Integrate-and-Fire (LIF) neuron model, where the membrane potential decays with constant $\alpha=0.9$, integrates incoming weighted spikes, and emits an output spike when the threshold $\theta=1.0$ is exceeded. Since spike activations are non-differentiable, surrogate gradient-based backpropagation is used during training with a Fast-Sigmoid surrogate function ($\beta=10$). Following convolution, 2×2 max pooling reduces spatial dimensionality while preserving salient activations. The temporally ordered spike features are fed into a Bidirectional GRU (BiGRU) for sequential modeling. At each time step r , the GRU computes:

$$b_r = \sigma(Z_b p_r + J_b q_{r-1} + f_b) \quad (18)$$

Where,

- b_r -Reset gate,
- σ - Sigmoid activation function,
- Z_b -input weight matrix for reset gate,
- p_r -Current input at time step r ,
- J_b -Recurrent weight matrix for reset gate,
- q_{r-1} -Hidden state from previous time step,
- f_b -Bias vector for reset gate.

$$d_r = \sigma(Z_d p_r + J_d q_{r-1} + f_d) \quad (19)$$

Where,

- d_r -Update Gate,
- Z_d -Input weight matrix for update gate,
- J_d -Recurrent weight matrix for update gate,
- f_d -Bias vector for update gate.

$$\tilde{q}_r = \tanh(Z_q p_r + J_q (q_{r-1} \otimes b_r) + f_r) \quad (20)$$

Where,

- $\tilde{q}_r$ -Candidate Hidden State,
- Z_q -Input weight matrix for candidate state,
- J_q -Recurrent weight matrix for candidate state,
- $q_{r-1} \otimes b_r$ - Elementwise multiplication of previous hidden state and reset gate,
- f_r -Bias vector for candidate state,
- $\tanh$ - Activation function to bound the output between $[-1, 1]$.

$$q_r = (1 - d_r) \otimes q_{r-1} + d_r \otimes \tilde{q}_r \quad (21)$$

Where,

- q_r -Final Hidden State,

- d_r -Controls the new candidate state,
- $\otimes$ -Elementwise multiplication operator.

The BiGRU consists of two parallel GRU layers that process the sequence forward and reverse directions. The two hidden state vectors are concatenated to form a complete representation of the input signal:

$$Q_r = [\vec{Q}_r \oplus \overleftarrow{Q}_r] \quad (22)$$

Where,

- Q_r -final hidden state vector at time step r ,
- $\vec{Q}_r$ -hidden state vector from the forward GRU,
- $\overleftarrow{Q}_r$ -hidden state vector from the backward GRU,
- $\oplus$ -concatenation.

This bidirectional modeling captures both prior and future contextual information for enhancing robustness in detecting RPW acoustic patterns. Finally, a fully connected layer with softmax activation performs binary classification between RPW and NRPW sounds.

Table 3 summarizes the key specifications of the SCNN–BiGRU for RPW detection. The model takes a 128×128 spectrogram as input and employs 3D convolutional layers with filters of 32, 64, and 128 using a $3 \times 3 \times 3$ kernel to extract spatial–temporal features. A spike-based representation with 10-time steps ($T=10$) is used to model temporal dynamics. The extracted features are processed through a Bidirectional GRU to capture forward and backward dependencies. A dropout rate of 0.5 and batch size of 32 are used for stable training. The network is trained for 100 epochs using the Adam optimizer with a learning rate of 0.001, and the Cross-Entropy loss function is employed to optimize binary classification performance. Finally, a softmax layer performs the RPW and NRPW classification.

3.6 Fuzzy Based Decision Making

The outputs obtained from the visual model and audio model are integrated using a Mamdani Fuzzy Inference System (FIS) to determine the overall health condition of the coconut tree. The final decision is communicated to farmers through the IoT Blynk application for timely intervention. Let P_{DT} represent the probability of Diseased Tree obtained from the image model and P_{RPW} represent the probability of RPW obtained from the audio model. Both outputs are normalized within the range $[0, 1]$.

Table 3 SCNN–BiGRU Training Hyperparameters

Component	Configuration
Input Size	128×128 Spectrogram
SCNN Layers	3D Convolution
Filters	32, 64, 128
Kernel Size	$3 \times 3 \times 3$
Spike Time Steps (T)	10
BiGRU	Bidirectional
Dropout	0.5
Batch Size	32
Learning Rate	0.001
Number of Epochs	100
Loss Function	Cross-Entropy Loss
Output Layer	Softmax

Table 4 Gaussian Membership Function Parameters for Fuzzification

Linguistic Term	Center (c)	σ
Low	0.2	0.1
Medium	0.5	0.1
High	0.8	0.1

Table 5 Fuzzy Rule Base for Coconut Tree Health

Rule	DT	RPW	Tree Condition	Output
1	High	High	Not Healthy	Alert
2	High	Low	Not Healthy	Alert
3	Low	High	Not Healthy	Alert
4	Low	Low	Healthy	No Alert

3.6.1 Fuzzification

Fuzzification is the initial stage of the Mamdani FIS, where the crisp probability values obtained from the visual model (P_{DT}) and audio model (P_{RPW}) are transformed into fuzzy linguistic variables using Gaussian membership functions.

$$\mu(x) = e^{-\frac{(x-c)^2}{2\sigma^2}} \quad (23)$$

Where,

- $\mu(x)$ - Membership degree of input value x in a fuzzy set,
- x - Crisp input probability value (either P_{DT} or P_{RPW}),
- c - Center of the Gaussian membership function,
- σ - Standard deviation of the Gaussian membership function.

Each input variable is represented using three linguistic terms to describe its probability level. These linguistic terms namely Low, Medium, and High capture the degree of certainty associated with the predicted probabilities. This representation enables the fuzzy inference system to handle uncertainty effectively and perform rule-based reasoning in a fuzzy domain before defuzzification.

Table 4 shows the Gaussian membership function parameters used to convert the crisp probabilities from the visual and audio models into linguistic variables during fuzzification. The input probabilities obtained from the visual (P_{DT}) and audio (P_{RPW}) models are mapped to linguistic terms using Gaussian membership functions. Each term Low, Medium, and High is defined by its center (c) and spread (σ). These membership functions allow the Mamdani FIS to evaluate tree health in a fuzzy domain before defuzzification produces a final crisp output.

3.6.2 Fuzzy Rule Base

The fuzzy rule base forms the core decision-making mechanism of the Mamdani FIS, where the fuzzified outputs from the visual and audio models are combined using predefined IF–THEN rules to determine the overall health condition of the coconut tree and generate an appropriate alert decision.

Table 5 shows the Mamdani fuzzy rule used to determine the overall health status of the coconut tree by combining outputs from the visual (P_{DT}) and audio (P_{RPW}) models. Each rule evaluates a combination of input probabilities to infer the tree condition. The fuzzy inference system applies the minimum operator for the logical AND operation that aggregates all activated rules using the maximum operator and converts the fuzzy output into a crisp value through centroid defuzzification. For example, if the visual model predicts a diseased tree probability of 0.75 and the audio model predicts a red palm weevil probability of 0.65, both values strongly belong

to the High linguistic term. Consequently, Rule 1 states “If the diseased tree probability is High and the red palm weevil probability is High then the tree condition is Not Healthy with an Alert” is activated. After aggregation and defuzzification, the resulting crisp output is approximately 0.72. Since this exceeds the predefined decision threshold of 0.6, the tree is classified as Not Healthy and an alert notification is sent to the farmer via the IoT Blynk application

Input: Stem images and Acoustic signals

Output: Classified Coconut Tree Health Status (Healthy / Not Healthy)

Step-1: Stem Image Pre-Processing

Step-1.1: Apply SavG filter to remove noise from stem images using equation (1)

Step-1.2: Resize the filtered image to 640×640 resolution

Step-1.3: Apply image augmentation techniques (rotation, horizontal flip, zoom) during training phase

Step-2: Acoustic Signal Pre-Processing

Step-2.1: Acquire acoustic signal using INMP441 microphone sensor

Step-2.2: Segment signal into overlapping frames using Hamming window equations (3) and (4)

Step-2.3: Perform STFT using 512-point FFT using equation (5)

Step-2.4: Generate magnitude spectrogram using equation (6)

Step-2.5: Normalize spectrogram using Min–Max normalization equation (7)

Step-2.6: Apply audio augmentation

Step-3: Coconut Stem Damage Detection using YOLOv9

Step-3.1: Extract backbone features using convolution and ResNeSt split-attention using equation (9) and (10)

Step-3.2: Fuse hierarchical features using FPN and PAN modules using equations (11) and (12)

Step-3.3: Compute objectness score using equation (13)

Step-3.4: Optimize detection using total loss function in equation (14)

Step-3.5: Classify stem image as DT or NDT

Step-4: RPW Sound Classification using SCNN-BiGRU

Step-4.1: Convert normalized spectrogram into spike-encoded 3D tensor using rate coding ($T = 10$ -time steps)

Step-4.2: Extract spike-based spatial–temporal features using Spiking CNN in equations (15) to (17)

Step-4.3: Model temporal dependencies using BiGRU equations (18) to (22)

Step-4.4: Apply Softmax activation to classify audio as RPW and NRPW

Step-4.5: Obtain RPW probability output P_{RPW}

Step-5: Multimodal Fusion using Mamdani Fuzzy System

Step-5.1: Convert detection probability P_{DT} and RPW probability P_{RPW} into fuzzy membership values using equation (23)

Step-5.2: Apply Mamdani fuzzy inference rules for health decision

Step-5.3: Defuzzify the output to obtain final tree health status

Step-6: IoT Notification

Step-6.1: If the final tree health status is Not Healthy, send an alert through the Blynk application.

Step-6.2: Otherwise, no alert is generated.

Algorithm 1 RED-BIYO for Multimodal RPW Detection Framework

4 Result and Discussion

The experimental study was conducted using the Spyder IDE within the Anaconda environment on a Windows 10 operating system. The system configuration included an Intel Core i7 processor, 16 GB RAM, and an NVIDIA RTX 3060 GPU for accelerated model training and inference. The proposed IoT-based RED-BIYO framework integrates acoustic signals from the INMP441 microphone sensor and stem images for accurate RPW detection

in coconut trees. The proposed RED-BIYO model was assessed with diverse performance metrics based on the gathered data. Experimental results of the RED-BIYO model for DT and NDT detection, RPW and NRPW detection are shown in Fig. 4.

The experimental outcomes of the proposed RED-BIYO model evaluated on four types of detection are illustrated in Fig. 4. Column 1 shows the environment data of coconut tree plantations representing various environmental contexts. Column 2 shows the input image/signal coconut stem image and corresponding acoustic signals captured as waveforms, representing the sound produced by Red Palm Weevils. In column 3 shows the pre-processing process the visual images are enhanced to highlight damaged regions, while the acoustic signals are transformed into spectrograms for detailed frequency analysis. Finally, the column 4 shows the detection the damaged stem and the acoustic analysis confirms the presence or absence of RPW.

Figure 5 shows the real time experimental setup. A farmer alert system is developed to access a real-time data acquisition from the coconut tree health notify send alert to farmer, if any abnormal situation is raised. The

Fig. 4 Experimental outcomes of the RED-BIYO

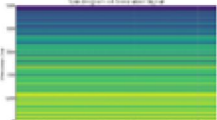
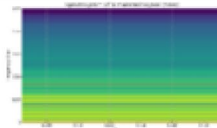
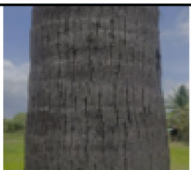
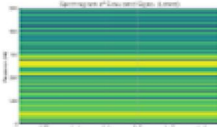
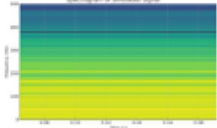
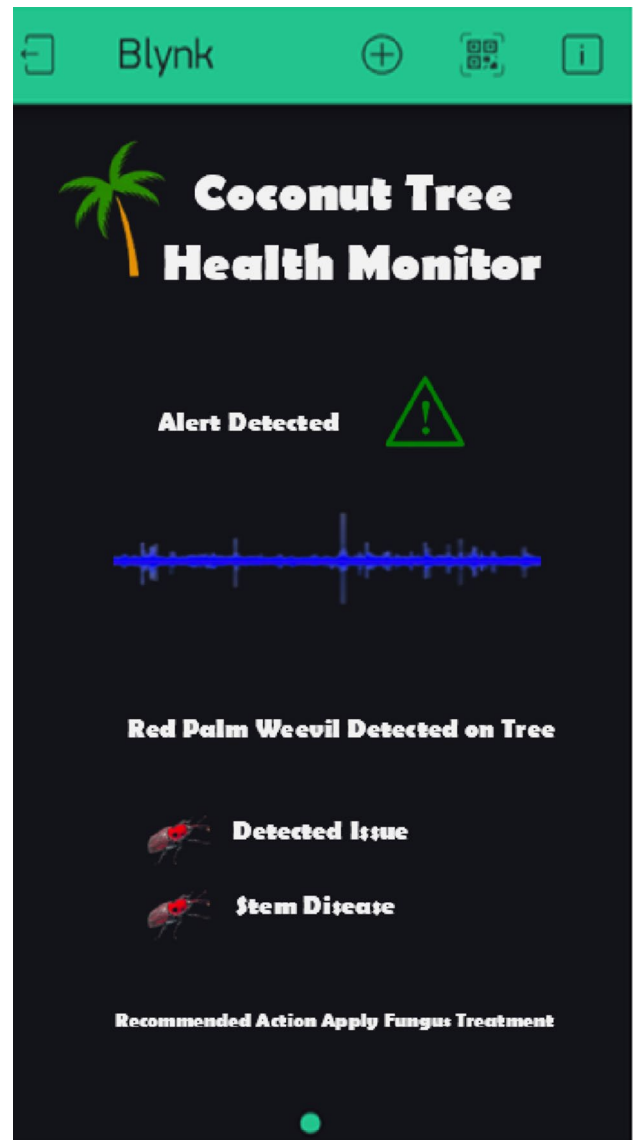
Environment data	Input image/ Signal	Pre-processing	Detection
			
			NRPW
			
			RPW
			NDT
			NRPW
			
			RPW

Fig. 5 Real-time experimental setup

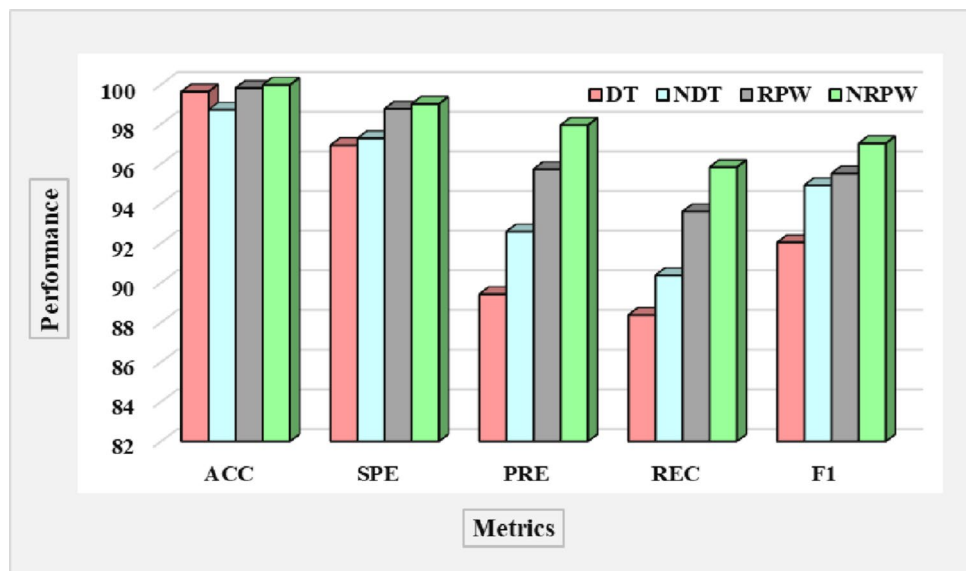


applied fuzzy rule makes recommendation to the farmer to check the status of the coconut field. Figure 6 displays the real time experimental setup of the RED-BIYO model. The system includes sensors mounted on the coconut tree to continuously monitor environmental and health parameters. All collected data is conducted wirelessly to a central server for real-time analysis and decision-making.

4.1 Materials and Methods

4.1.1 Dataset Description

The proposed study utilizes a multimodal dataset comprising stem images and acoustic recordings collected from coconut plantations in Tamil Nadu, India, between June 2024 and December 2024. The dataset includes labeled samples representing diseased and non-diseased conditions in both visual and acoustic modalities. The stem images capture visible external symptoms, while the acoustic recordings reflect internal infestation activity. For experimental evaluation, a five-fold cross-validation strategy was employed to ensure robustness and generalization. In each fold, 80% of the data were used for training and 10% for testing and 10% for validation while

Fig. 6 RED-BIYO classification performance across four classes**Table 6** Description of the Collected Multimodal Dataset

Category	Stem Image Dataset	Acoustic Dataset
Total Samples	3,000	2,400
Classes	2 (DT, NDT)	2 (RPW, NRPW)
Class Distribution	1,500 DT 1,500 NDT	1,200 RPW 1,200 NRPW
Resolution/Sampling	1024 × 768 pixels	16 kHz
Duration	-	10–30 s
Sensor Used	-	INMP441 Digital Microphone
Annotation	Expert annotated	Physical inspection + Expert validation
Collection Period	June 2024 – December 2024	
Location	Tamil Nadu, India	

maintaining class balance and preventing data leakage between folds. A fixed random seed (seed=42) was used during data splitting and model training to ensure reproducibility of the results. The detailed specifications of the collected multimodal dataset are summarized in Table 6.

The evaluation of the proposed RED-BIYO model including accuracy (ACC), precision (PRE), specificity (SPE), recall (REC), and F1-score (F1) are utilized to determine the efficacy of the proposed RED-BIYO model for detecting the DT and NDT.

$$ACC = \frac{P_{true} + N_{true}}{Total\ no.of\ samples} \quad (24)$$

$$SPE = \frac{N_{true}}{N_{true} + P_{false}} \quad (25)$$

$$PRE = \frac{P_{true}}{P_{true} + P_{false}} \quad (26)$$

$$REC = \frac{P_{true}}{P_{true} + N_{false}} \quad (27)$$

Table 7 Performance analysis of the RED-BIYO model

Class	ACC	SPE	PRE	REC	F1	MCC
DT	99.64	96.93	89.42	88.37	92.04	98.78
NDT	98.73	97.28	92.58	90.37	94.92	97.45
RPW	99.82	98.77	95.73	93.59	95.51	98.56
NRPW	99.98	99.02	97.95	95.83	97.03	96.94

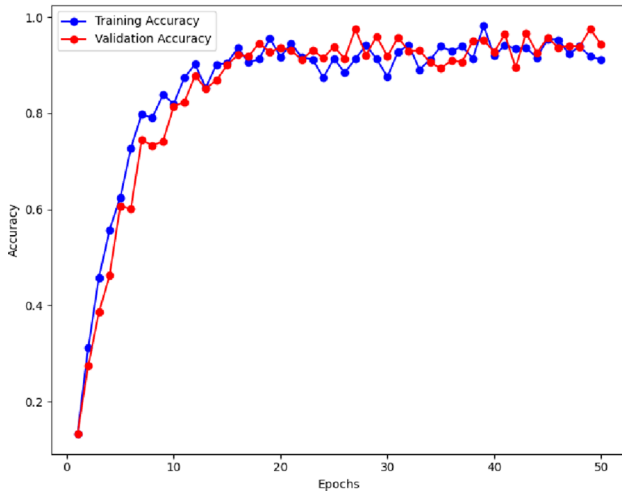


Fig. 7 Accuracy curve of the RED-BIYO

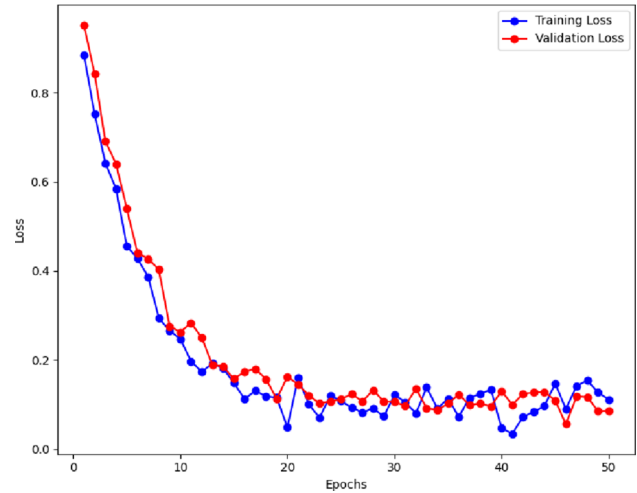


Fig. 8 Loss curve of the RED-BIYO

$$F1 = 2 \left(\frac{PRE * REC}{PRE + REC} \right) \quad (28)$$

Where,

- N_{true} -true-positives,
- N_{false} - true-negatives,
- P_{true} - false negatives,
- P_{false} -false positives.

Table 7 displays the classification performance attained utilizing the RED-BIYO for detecting RPW in coconut trees. The RED-BIYO model also achieves SPE of 98%, PRE of 93.92%, REC of 92.04% MCC of 97.93% and the F1 of 94.87%. Figure 8, displays the graphical evaluation of the RED-BIYO model. The performance analysis of the RED-BIYO model achieves an overall ACC of 99.54% based on Coconut Tree Disease database.

Figure 7 signifies the accuracy curve, in which accuracy and epochs are placed in opposing axes, as the number of epochs and model accuracy improve. Figure 8 denotes the epoch versus loss curve in which the loss of the model reduces as the number of epochs increases. The RED-BIYO model achieves an ACC of 99.54% based on gathered dataset.

Figure 9 represents the classification performance of the DL-based RED-BIYO model to detect RPW in coconut trees using microphone sensor and stem image across four classes DT, NDT, RPW and NRPW. The diagonal values of 0.99, 0.98, 0.99 and 0.99 indicate high accuracy in classifying each category.

Figure 10 illustrates the ROC curves of the proposed RED-BIYO model for the four classes: DT, NDT, RPW, and NRPW. The ROC curves demonstrate the trade-off between the TPR and FPR across different threshold

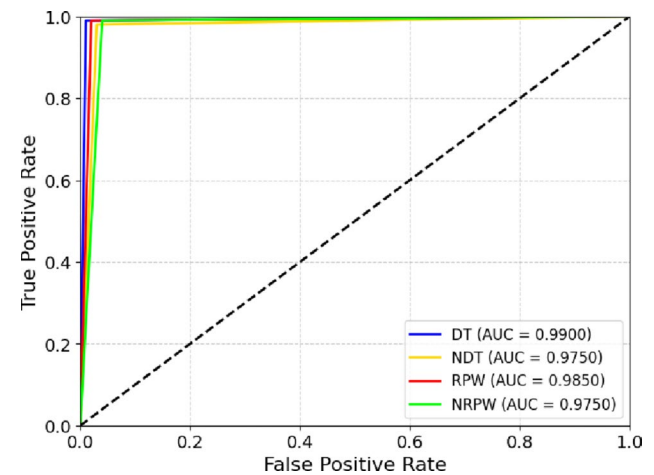
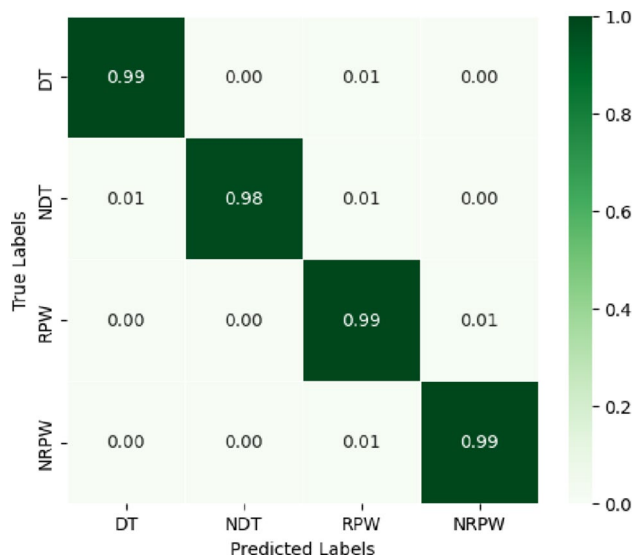


Fig. 10 ROC curve of the proposed RED-BIYO

Fig. 9 Confusion matrix of the RED-BIYO

values. The high AUC values observed for all classes indicate excellent discriminative capability of the model. Among the four classes, NRPW and RPW exhibit the highest classification performance, followed closely by DT and NDT. AUC values confirm that the proposed RED-BIYO model achieves strong class separability with minimal misclassification. Overall, the ROC analysis validates the robustness and reliability of the proposed framework under the gathered dataset.

Figure 11 presents the detection rate performance of the YOLOv9 model compared with SSD [25], YOLOv4 [26], YOLOv5 [27], and YOLOv7 [28]. The results show that YOLOv9 achieves the highest detection rate for demonstrating its superior accuracy and reliability over existing method.

Table 8 illustrates the five-fold cross-validation performance of the proposed RED-BIYO model. The dataset was partitioned into five mutually exclusive folds to evaluate the robustness, stability, and generalization capability of the framework while maintaining balanced class distribution across each fold for unbiased assessment. The experimental results demonstrate consistent and reliable performance across all folds. The RED-BIYO model achieves an average ACC of $97.92 \pm 1.63\%$, SPE of $96.30 \pm 1.44\%$, PRE of $95.46 \pm 0.94\%$, REC of $93.89 \pm 1.24\%$, and F1 of $94.26 \pm 0.93\%$. The relatively low standard deviation values indicate stable learning behaviour and confirm that the model does not suffer from overfitting for ensuring strong generalization performance on unseen data.

4.2 Comparative Analysis

In this comparative analysis, the RED-BIYO model was related with various existing methods based on dissimilar parameters. The RED-BIYO was compared with traditional networks namely LSTM, BiGRU, BiLSTM and CNN-BiGRU and proposed SCNN-BiGRU for the detection of RPW in coconut trees.

According to Table 9, the proposed SCNN-BiGRU is higher than that of the classic networks like LSTM, BiGRU, BiLSTM and CNN-BiGRU. The proposed SCNN-BiGRU improves its accuracy by 2.22%, 4.95%, 3.43% and 1.41% better than LSTM, BiGRU, BiLSTM and CNN-BiGRU respectively. This significant improvement demonstrates the detection capabilities of the SCNN-BiGRU model over traditional DL architectures.

Figure 12 illustrates the sample output for defect detection with existing techniques like Faster RCNN, SSD and YOLOv6 with the YOLOv9 model. The proposed approach achieves higher accuracy and efficiency than other

Fig. 11 Detection rate comparison of the proposed YOLOv9 model vs. existing techniques

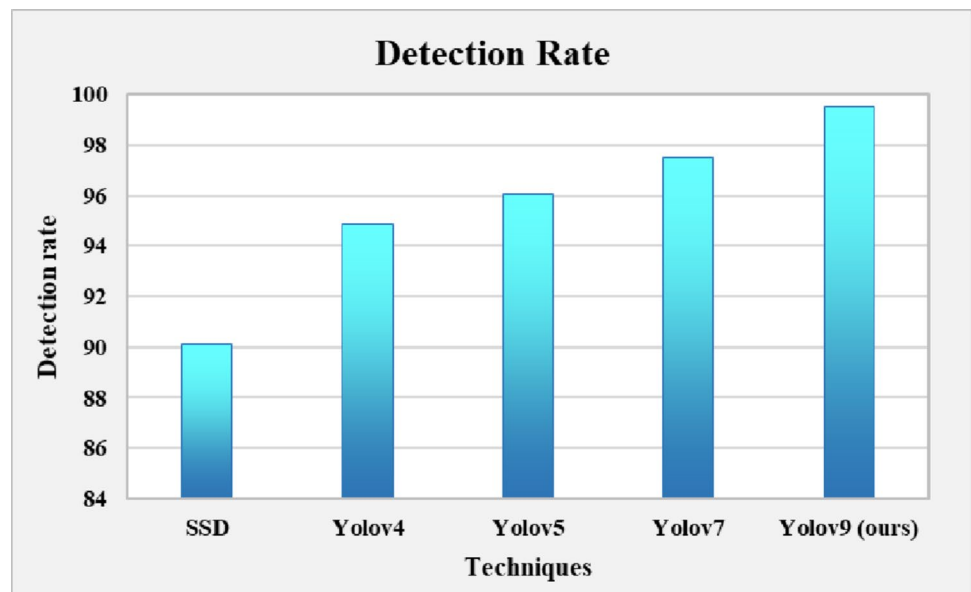


Table 8 Five-Fold cross-validation performance of RED-BIYO

Fold	ACC	SPE	PRE	REC	F1
Fold-1	98.01	97.01	95.46	95.30	94.38
Fold-2	95.34	94.45	96.36	94.65	93.10
Fold-3	99.03	96.85	96.21	93.39	95.37
Fold-4	97.70	95.20	95.35	94.10	93.58
Fold-5	99.54	98.00	93.92	92.04	94.87
Average (Mean±SD)	97.92±1.63	96.30±1.44	95.46±0.94	93.89±1.24	94.26±0.93

Table 9 Comparison of existing models and SCNN-BiGRU model

Techniques	ACC	SPE	PRE	REC	F1
LSTM [29]	97.37	96.84	88.21	86.26	88.53
BiGRU [30]	94.84	94.59	91.03	91.37	86.38
BiLSTM [31]	96.23	93.62	89.61	88.83	89.42
CNN-BiGRU [32]	98.15	95.07	90.49	91.18	90.74
SCNN-BiGRU (Ours)	99.54	98.00	93.92	92.04	94.87

object detection methods. YOLOv9 demonstrates improved localization precision and reduced false positives, particularly in complex backgrounds, challenging textures, varying lighting conditions.

According to Table 10, the proposed YOLOv9 outperforms traditional detection networks such as SSD, Faster RCNN, Mask RCNN, and YOLOv6. The proposed YOLOv9 achieves accuracy improvements of 7.21%, 6.46%, 4.54%, and 1.85% over SSD, Faster RCNN, Mask RCNN, and YOLOv6. These results indicate that YOLOv9 delivers superior performance compared to the existing models based on the defined assessment metrics.

Table 11 shows the efficiency of the RED-BIYO for detecting RPW in coconut trees using utilizing different datasets. The existing datasets namely Kaggle RPW, Date Palm RPW and Coconut Pest and Disease dataset establish low accuracy range. The proposed RED-BIYO model significantly achieves 99.54% AC, 93.92% PR and 92.04% RE in the Coconut Tree Disease Dataset. The outcomes emphasize the RED-BIYO model attains better results in the gathered Coconut Tree Disease Dataset than the other datasets.

Table 12 shows the comparison of different existing models with the proposed RED-BIYO model based on the gathered dataset. The RED-BIYO improves the overall accuracy by 2.75%, 9.38%, 2.61%, 22.16%, 8.54%, and 4.39% over YOLOv5, ResNet50, Mask R-CNN, InceptionResNetV2, YOLOv3, and MIN-SVM models,

Fig. 12 Comparison of existing models with a proposed Yolov9 for DT and NDT detection

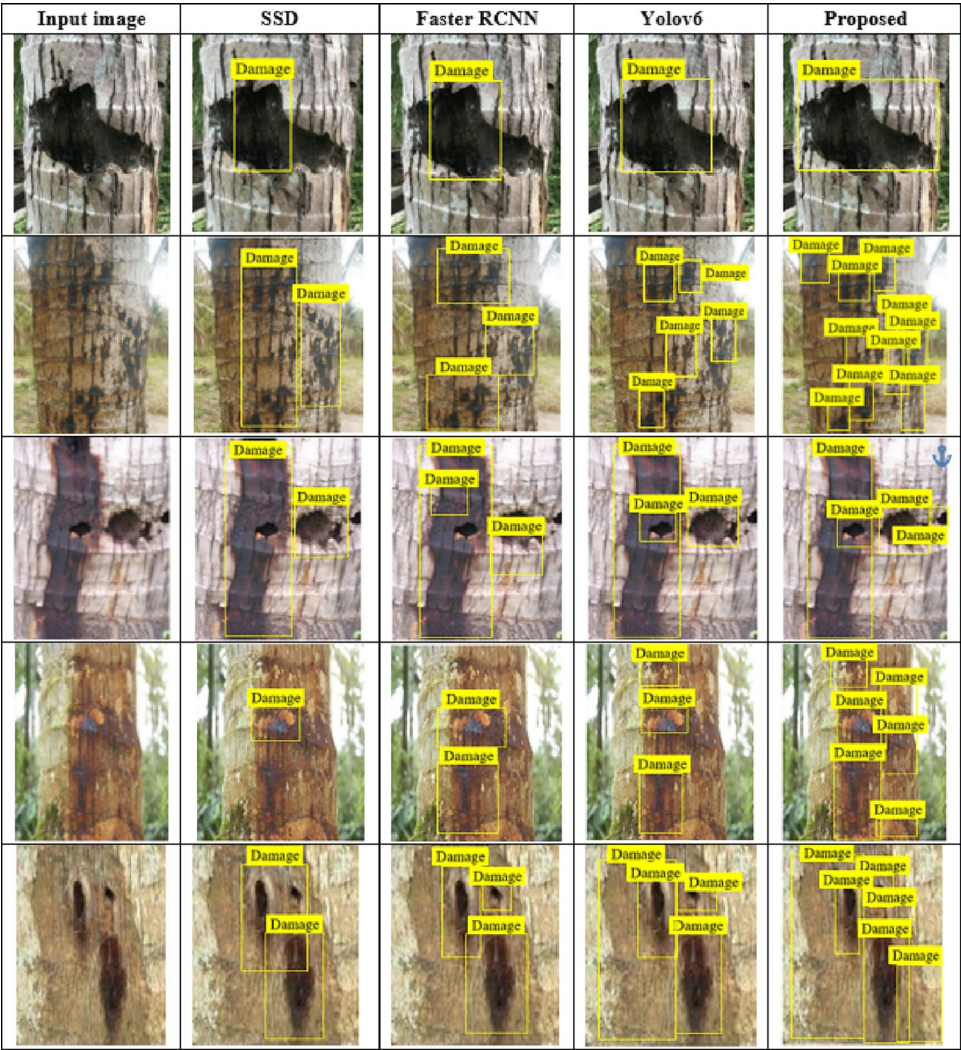


Table 10 Comparison of existing models and Yolov9 model

Networks	ACC	SPE	PRE	REC	F1
SSD [25]	92.36	90.37	88.27	79.83	87.84
Faster RCNN [33]	93.10	91.93	89.02	81.27	89.27
Mask RCNN [34]	95.02	93.18	90.28	84.20	91.73
Yolov6 [35]	97.69	95.01	91.48	89.98	92.14
Yolov9 (Ours)	99.54	98.00	93.92	92.04	94.87

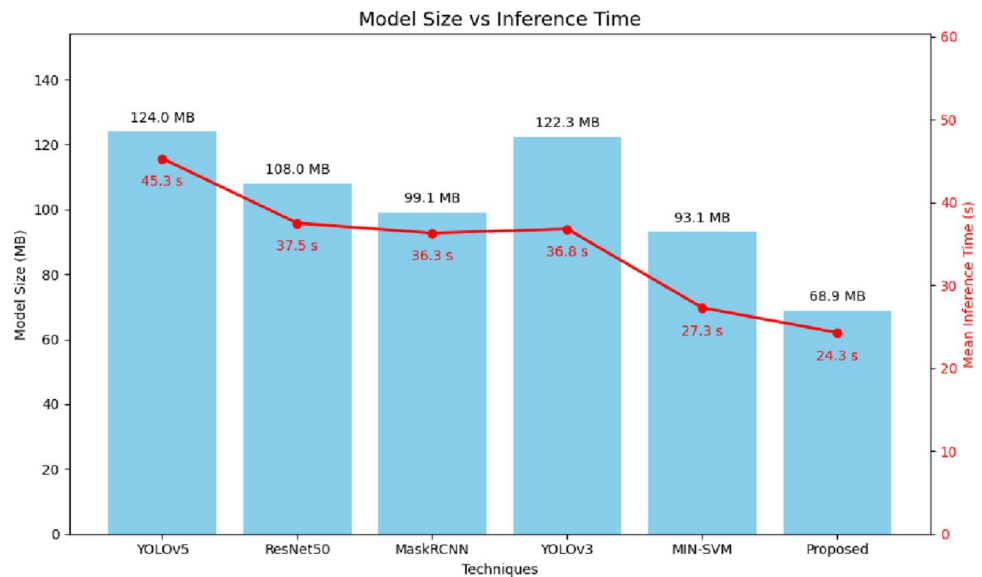
Table 11 Dataset comparison of the proposed RED-BIYO model

Dataset	ACC	PRE	REC
Kaggle RPW	95.38	93.72	94.51
Date Palm RPW	96.74	94.83	95.21
Coconut Pest and Disease dataset	94.15	92.20	93.03
Coconut Tree Disease Dataset (ours)	99.54	93.92	92.04

Table 12 Accuracy comparison of proposed RED-BIYO vs. existing models

Author	Techniques	ACC	<i>p</i> -value
Wijethunga, C.D., et al., [15]	YOLOv5	96.87%	0.054
Iqbal, M.S., et al., [16]	ResNet50	91%	0.074
Vidhanaarachchi, S., et al., [17]	Mask R-CNN	97%	0.027
Singh, P., et al., [19]	InceptionResNetV2	81.48%	0.094
Karunaratna, et al., [20]	YOLOv3	91.7%	0.048
Megalingam, R.K., et al., [24]	MIN-SVM	95.35%	0.029
Proposed	RED-BIYO	99.54%	0.007

Fig. 13 Model size and inference time comparison across detection networks



respectively. From this comparative analysis, the RED-BIYO model achieves superior accuracy and demonstrates improved statistical reliability in coconut tree disease detection. YOLOv5 [15] achieved 96.87% accuracy but is limited in capturing deeper contextual features, while ResNet50 [16] achieved 91% accuracy and is mainly suitable for classification tasks. The proposed RED-BIYO outperforms all existing models with an accuracy of 99.54% due to enhanced multimodal feature extraction and fusion capability. The proposed RED-BIYO model achieved a *p*-value of 0.007, indicating that the improvement in accuracy over the existing models is statistically significant.

Figure 13 compares the model size (MB) and average inference time (s) of existing detection networks, including YOLOv5, ResNet50, Mask R-CNN, InceptionResNetV2, YOLOv3, MIN-SVM, with the proposed RED-BIYO. The inference time was calculated as the average processing time per sample over the test set with a batch size of 1. All models were evaluated under identical conditions using an Intel Core i7 processor, 16 GB RAM, and an NVIDIA RTX 3060 GPU in the PyTorch framework. The model size refers to the final trained weight file without compression. The RED-BIYO achieved the lowest inference time of 24.310 s and the smallest model size of 68.9 MB for demonstrating improved computational efficiency under the experimental setup.

4.3 Ablation Study

In this analysis, the proposed RED-BIYO model was evaluated with and without stem image and Audio signals for detecting RPW in coconut trees.

Table 13 displays the ablation results of the RED-BIYO using stem images, audio signals, and their combination. Using only stem images yields 98.12% accuracy, while audio alone gives 97.05%. The combination of both achieves the highest accuracy of 99.54%. Additionally, the F1 score and precision also improve with multimodal

Table 13 Ablation study of the RED-BIYO model

Metrics	Stem image	Audio signals	With stem image and Audio signals
Accuracy	98.12	97.05	99.54
F1 score	93.86	92.93	94.87
Precision	92.45	90.56	93.92

input achieving 94.87% and 93.92%, respectively. These results highlight the efficacy of feature fusion in enhancing overall proposed model performance.

5 Conclusion

This paper introduces a novel DL-based RED-BIYO for detecting RPW in coconut trees utilizing multimodal data. The stem images are pre-processed by SavG filter to remove the noise from the images. The acoustic signals are pre-processed using Hamming window-based segmentation and STFT to generate normalized spectrogram representations. The YOLOv9 model leverages a ResNeSt-based split-attention backbone to extract multi-scale features from noise free images, enabling precise detection of DT and NDT. Simultaneously, the spectrograms processed by SCNN-BiGRU for RPW detection. SCNN extracts spatial and frequency features, while BiGRU captures temporal patterns in both forward and backward directions. The multi model data are fused in fuzzy rules to assess coconut tree health and notify farmers via an IoT-enabled Blynk app. The proposed SCNN-BiGRU improves its accuracy by 2.22%, 4.95%, 3.43% and 1.41% better than LSTM, BiGRU, BiLSTM and CNN-BiGRU respectively. The proposed Yolov9 achieves accuracy improvements of 7.21%, 6.46%, 4.54%, and 1.85% over SSD, Faster RCNN, Mask RCNN, and Yolov6 respectively. The RED-BIYO model increases the overall accuracy by 2.75%, 9.38%, 2.61%, and 22.16% of YOLOv5, ResNet50, Mask R-CNN and InceptionResNetV2 model respectively.

5.1 Limitations and Future Work

The proposed RED-BIYO system requires moderate computational resources, limiting its deployment on low-power IoT devices for large-scale and real-time monitoring. In future the proposed RED-BIYO will be optimized for energy-efficient edge deployment to enable large-scale, continuous RPW monitoring in agricultural fields.

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Author Contributions The authors confirm contribution to the paper as follows: Study conception and design: Siron Anita Susan T, Manjunathan Alagarsamy, Sumathi Gurusamy; Data collection: Alima Beevi Azizur Rahuman, Ramya V, Sowmya Sundari L K; Analysis and interpretation of results: Siron Anita Susan T; Draft manuscript preparation: Manjunathan Alagarsamy, Sumathi Gurusamy, Sowmya Sundari L K. All authors reviewed the results and approved the final version of the manuscript.

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Data Availability The dataset used in this study is proprietary, collected from privately owned coconut plantations under field monitoring agreements. Due to privacy policies and the sensitive nature of the agricultural field data, the dataset cannot be publicly released.

Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval The data used in this study were collected from coconut plantations in Tamil Nadu, India, with prior permission from the respective landowners. The study does not involve human participants or animal experimentation therefore, Institutional Ethics Committee approval was not required. The official approval document will be provided during the thesis submission stage.

Human and Animal Rights This article does not contain any studies with human or animal subjects performed by any of the authors.

Informed Consent I certify that I have explained the nature and purpose of this study to the above-named individual, and I have discussed the potential benefits of this study participation. The questions the individual had about this study have been answered, and we will always be available to address future questions.

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Authors and Affiliations

**Siron Anita Susan T¹ · Manjunathan Alagarsamy² · Sumathi Gurusamy³ ·
Alima Beevi Azizur Rahuman⁴ · V. Ramya⁵ · L. K Sowmya Sundari⁶**

✉ L. K Sowmya Sundari
sowmyasundari5qq@outlook.com

Siron Anita Susan T
anitasusan.t@gmail.com

Manjunathan Alagarsamy
manjunathankrct@gmail.com

Sumathi Gurusamy
sumathigurusamy@gmail.com

Alima Beevi Azizur Rahuman
mail2alimabeevi@gmail.com

V. Ramya
ramyavojjala55@gmail.com

- ¹ Department of Computer Science and Engineering, SRM Institute of Science and Technology, Tiruchirappalli 621105, Tamil Nadu, India
- ² Department of Electronics and Communication Engineering, K.Ramakrishnan College of Technology, Trichy 621112, Tamil Nadu, India
- ³ Department of Computer Science and Information Technology, Kalasalingam Academy of Research and Education, Krishnankoil 626126, Tamil Nadu, India
- ⁴ Department of Computer Science and Engineering, Panimalar Engineering College, Chennai 600123, Tamil Nadu, India
- ⁵ Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Bowrampet, Hyderabad 500043, Telangana, India
- ⁶ School of Computing and Information Technology, REVA University, Bangalore 560064, Karnataka, India